

Amirhosein Azarpour

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[LinkedIn](#) | [GitHub](#) | [Google Scholar](#) | [ResearchGate](#)

RESEARCH INTERESTS

Deep Learning · Machine Learning · Medical AI · Trustworthy/Efficient AI · Computer Vision · Time-Series Analysis

EDUCATION

Shahid Beheshti University

Sep 2025 – Current

Master's Degree, Data Mining — Tehran, Iran

- GPA: ~18.5/20 — Supervisor: [Dr. Kheradpisheh](#)

Amirkabir University of Technology (Tehran Polytechnic)

Sep 2021 – Sep 2025

Bachelor's Degree, Computer Science — Tehran, Iran

- GPA: ~16.7/20 (Last Two Years: 18/20)
- Thesis: ORB-SVM: An Innovative Hybrid Framework for Efficient Brain Tumor Detection from MRI Scans (Supervisor: [Dr. Shamsi](#))

PUBLICATIONS & PREPRINTS

1. **Azarpour, A.** ORB-SVM: An Innovative Hybrid Framework for Efficient Brain Tumor Detection from MRI Scans. IEEE, 2025. Status: Published. [IEEE Xplore](#)
2. Kazemi, S. M., **Azarpour, A.**, [Riahi-Madvar, M.](#) CRNN-KAN: A Lightweight Hybrid Framework for Robust Phonocardiogram Classification. Informatics in Medicine Unlocked. Status: Under Review.
3. **Azarpour, A.**, Kazemi, S. M. SW-KAN: Kolmogorov-Arnold Networks with Stieltjes-Wigert q-Orthogonal Polynomials. Applied Mathematics and Computation. Status: Under Review.
4. **Azarpour, A.** RecKAN: Kolmogorov-Arnold Networks with a Learnable Recursive Polynomial Basis. arXiv preprint, 2026. [arXiv](#)
5. **Azarpour, A.**, [Kheradpisheh, S. R.](#) From S4 to Mamba: A Unified Survey of Modern State Space Models. Manuscript in preparation, 2026.

RESEARCH EXPERIENCE

Research Assistant

Mar 2026 – Aug 2026

Supervisor: [Dr. Riahi-Madvar](#) — Shahid Beheshti University, Tehran, Iran

Lightweight Hybrid Deep Learning for Medical Signal Classification (CRNN-KAN)

- Proposed CRNN-KAN, integrating a CNN-BiLSTM backbone with an EfficientKAN head (SiLU + sinusoidal bases) for robust phonocardiogram (PCG) classification.
- Achieved competitive AUC scores across PhysioNet, CirCor, and PASCAL datasets while reducing trainable parameters to 1.31M (100× reduction vs. VGG16); validated interpretability via Score-CAM.

Research Assistant

2024 – 2025

Supervisor: [Dr. Shamsi](#) — Amirkabir University of Technology, Tehran, Iran

Efficient Brain Tumor Detection from MRI (ORB-SVM)

- Designed a hybrid ORB feature extraction + SVM classification pipeline, reducing input dimensionality by ~99.5% while retaining diagnostically relevant patterns.
- Achieved 97.5% classification accuracy on the Br35H dataset, offering a resource-efficient alternative to large deep learning architectures.

Research Assistant

Apr 2026 – Present

Supervisor: [Dr. Kheradpisheh](#) — Shahid Beheshti University, Tehran, Iran

- Co-authored the survey “**From S4 to Mamba: A Unified Survey of Modern State Space Models**” with [Dr. Kheradpisheh](#), covering S4, HiPPO, selective SSMs, and the Mamba/Mamba-2 architectures.
- Continuing research on Mamba-based selective state space models, extending the survey’s findings toward new architectural and application directions.

Learnable Polynomial Basis Functions for KANs (RecKAN)

- Developed RecKAN, defining the KAN basis via a learnable second-order polynomial recurrence that recovers classical families (Chebyshev, Fibonacci, Pell, Jacobsthal); outperforms parameter-matched KAN baselines on image, text, and biomedical time-series classification.

Independent Researcher

Jan 2026 – Aug 2026

Collaborative Research — Tehran, Iran

Stieltjes–Wigert q -Orthogonal Polynomial KANs (SW-KAN)

- Developed SW-KAN, replacing B-splines with Stieltjes-Wigert q -orthogonal polynomials via a smooth exponential-of-tanh domain mapping and an $O(N)$ three-term recurrence, outperforming 18 baseline polynomial KAN variants.

TEACHING & ACADEMIC EXPERIENCE

- Teaching Assistant: Principles of Software Design & Workshop (Instructor: [Dr. Ghorbanali](#))
- Teaching Assistant: Software Mathematics (Instructor: [Dr. Shamsi](#))
- Teaching Assistant: Image Processing / Computer Graphics (Instructor: [Dr. Shamsi](#))
- Teaching Assistant: Artificial Intelligence (Instructor: [Dr. Ghatee](#))
- Head TA: Machine Learning (Instructor: [Dr. Riahi-Madvar](#))
- Teaching Assistant: Deep Learning (Instructor: [Dr. Hajimohammadi](#))

SELECTED TECHNICAL PROJECTS

- **Behavioral Modeling & Recommendation Systems:** End-to-end Instacart recommendation pipeline featuring behavioral EDA, user/product/user-product feature engineering, PCA/t-SNE/UMAP analysis, and LightGBM/XGBoost reorder prediction evaluated with ROC-AUC and Top- K ranking metrics. [\[GitHub\]](#)
- **Data Mining & Deep Learning Project Handbook:** Collection of reproducible deep learning projects spanning leakage-aware ECG anomaly detection, PPO-based GPT-2 sentiment steering and reward-hacking analysis, multivariate weather forecasting with recurrent and attention models, Transformer data-efficiency evaluation, and VAE latent-space attribute manipulation. [\[GitHub\]](#)
- **RecKAN:** Learnable recursive polynomial-basis KAN framework, generalizing Chebyshev, Fibonacci, Pell, and Jacobsthal-type recurrences. [\[GitHub\]](#)
- **SW-KAN:** Stieltjes–Wigert q -orthogonal polynomial KAN implementation with learnable q , stable domain mapping, and MNIST/Fashion-MNIST/function-approximation experiments. [\[GitHub\]](#)

TECHNICAL SKILLS

- **Programming:** Python, C/C++, Java, R, MATLAB
- **DL Frameworks:** PyTorch, TensorFlow, Keras, NumPy, Pandas, Scikit-learn, OpenCV
- **Architectures & Specializations:** KAN, Mamba, SSM, LSTM, ViT, Deep RL, Computer Vision, Medical Image/Signal Processing, Time-Series Analysis, XAI (Score-CAM, Grad-CAM)
- **Tools & Platforms:** Git, CUDA, Jupyter, LaTeX
- **Languages:** Persian (Native), English (Fluent – Preparing for TOEFL)

COURSEWORK GRADES

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| • Artificial Neural Networks: 20 | • Principles of Computer Systems: 18.5 | • Numerical Linear Algebra: 19.2 |
| • Fundamentals of Reinforcement Learning: 18.5 | • Cryptographic Protocols: 18.5 | • Principles of Operating Systems: 18.13 |
| • Advanced Machine Learning: 18.5 | • Computer Networks: 19.55 | • Artificial Intelligence: 19.01 |

HONORS & AWARDS

- Top 0.1% Nationwide: Among ~150,000 applicants in the Iranian University Entrance Exam for B.Sc. in Computer Science, 2021.
- Qualified for the Final Round of the National Olympiad in Mathematics, National Organization for Development of Exceptional Talents (Sampad), 2018.
- Member of the National Organization for Development of Exceptional Talents (Sampad) — gifted students’ program during secondary education.

VOLUNTEERING & EXTRACURRICULAR ACTIVITIES

Executive Committee Member — AUT TRADE CHALLENGE EVENT

Sep 2023 – Sep 2024

Supervisor: [Dr. Mohades](#) — Amirkabir University of Technology